

List of past PC5203 lecture recordings (Jan – Apr 2021)

Week 1, crystal symmetry, point and space groups, Bloch's theorem.	12 Jan	https://nus-sg.zoom.us/rec/share/-7yixUayaYDQgJKnw1PdRGeUOYw83UNkJDCSI8KhtBIY-LdaNBiA0MIYIIVpJqE4.Qaobx4Ogj4A8p8TD Access Passcode: 8X..#8ja
	14 Jan	https://nus-sg.zoom.us/rec/share/B9DxJt5IgfPITL-ThxwD7R6BgGKwN8bLyAtGQfe16wLvlg5Xopqy3y7_hs1oYbpG.TJXjC1m5EbdIWGf7 Access Passcode: r70^?eIj
Week 2, lattice vibrations, ladder operators, phonon dispersion from DFT.	19 Jan	https://nus-sg.zoom.us/rec/share/U1wl0ZJ_0lixwNADnx5nFW_09Z-JZ0Xz8Hxqc7iWnT52FkUkPB78Q2aP_sinB7bd.PFODR_HlwSR3eDeu Access Passcode: 1eA2&2q!
	21 Jan	https://nus-sg.zoom.us/rec/share/32El1AK3r6cX1EYnBxrWs22Zd1zmDY_9lxQY4Se2sguKfJqJm0Om0SIMGGJOOWDF.nw2YTsWplBNbqLnv Access Passcode: bhfF%66#
Week 3, second quantization, electrons, graphene tight-binding model, band structures.	26 Jan	https://nus-sg.zoom.us/rec/share/Pbe59cAzlaoTV3sezUsz8BtbguPHOE7AkCS7u5cDWPZ1ONyzPhiQZPEESovJaGjX.FIldIP-nHhdmXI13 Access Passcode: xN+2fZ9P
	28 Jan	https://nus-sg.zoom.us/rec/share/7J2E9aldWbzTIWpVM39IOFgvVPOvQwHtZXQYimlCqgxg3fgRSrEReZWY2v2Gose-.FGUjtaOojxN98yVQ Access Passcode: gvus9i?S
Week 4, electron-electron interaction, electron-phonon interaction, Hartree-Fock method.	2 Feb	https://nus-sg.zoom.us/rec/share/eQWYf6rGtVR-fxj6Z-4suDsZ7VULkDMI6QGN6pcF3tIhUTT1y8gDxKAcBx2dBSiL.OVh5r9aaNmjO9LIQ Access Passcode: V4w2gL+P
	4 Feb	https://nus-sg.zoom.us/rec/share/D6QxzoukGvvFGfRb0ebxnpTAz6mq-e0Fyv-3fi7LUCbM28OvUBuCP6d7MnLg03qc.7SOfT4p2-xtD90Vr Access Passcode: WC8My^8d
Week 5, many-body theory, Dyson expansion, Green's functions.	9 Feb	https://nus-sg.zoom.us/rec/share/eFQ2z0-Cts8KUI4aLaCx8nxz3RTNYbJK6CfTiG_FVBPF05M4ckiC74sp3AsqC2Bn.MVmtpb-EH-3F9CgV Access Passcode: s@^9Mo0^

Week 6, many-body theory, Hedin equations, GW method.	16 Feb	https://nus-sg.zoom.us/rec/share/FAeHFJCAgalazDY_rJTe_oALDv8lZ1ZeTasoHfB8rk-OU77LRpSbsQlanOAMVeNf.Caw5zCr7IU-7MDY- Access Passcode: @e\$TBR0s
	18 Feb	https://nus-sg.zoom.us/rec/share/bozdOe0la5CfjoWE68MpQl1BPdUoHUHsKWCF-QW45xXpwNgRDZlcel6F64Tdyf4E.iILlWcWHvjLVVQjg Access Passcode: ^02\$pg82

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Week 7, derivation of Hedin equations.	2 Mar	https://nus-sg.zoom.us/rec/share/PJ7zin6rW9s_klqteUxG5oJsIYG4xi3NTza7vV0HJC1a7pVIEmlY2i9P4GplG4Cj.GP19yyBgBwNLYWeU Access Passcode: x5LkFV=6
Midterm test	4 Mar	
Week 8, electron-phonon interaction and SSH model, Peierls instability	9 Mar	https://nus-sg.zoom.us/rec/share/ipBkv1QNlzz9iwY-AyEfb1xfPMe08mKEds66tKXI_CtX5wGQBxdO5tuq3Fu.55qDOvs-LwtuZuv0 Access Passcode: mE0?8OYQ
	11 Mar	https://nus-sg.zoom.us/rec/share/s47G1Gv-1durdMHqSsgYUe0hBhEif6U6Kdk-QWy6RPkG0TwWS0XnTn2zeVgFmWv_.GIVWWPqx_SZlIhPE Access Passcode: *I+MH5U9
Week 9, electron transport, Landauer approach, thermoelectric effect.	16 Mar	https://nus-sg.zoom.us/rec/share/Gnx4qXVXYpMh6hxxj65Dk-cdui3egOKobxwUJa31DgLSrkEpzDleaRAQyxD_VIN9.jV4VhfJFwLb_flfc Access Passcode: .33HFd%W
	18 Mar	https://nus-sg.zoom.us/rec/share/qvAXB-AboTuBXkkudHF6jJzHiKyCbpoJ4G4Bc4e6x9a3KlXwT4_PJts9SJTtYB-F.7TdZCYsoHHY9nxNj Access Passcode: SW6kxG&3
Week 10, electron transport, Boltzmann (and	23 Mar	https://nus-sg.zoom.us/rec/share/XzNR36iU8ygii7QFKEyTg1GpbRk1ixnQ2gZ

Kubo) approach		2aMwkMafPPoXc-SFzZ-trMpQgF-H7.XwKEyQZ7Rxit8cBq Access Passcode: yVR3#%ed
	25 Mar	https://nus-sg.zoom.us/rec/share/PY2gaWePFb-d-izMcFnfXjG166LluQo-MZDr3pL4hSF_8L5D42tvW7-yh_bIA08Z.naRhTNqJ9TGzyV_h Access Passcode: um?kg3F?
Week 11, Berry phases in SSH model.	30 Mar	https://nus-sg.zoom.us/rec/share/yeYui3-4XiefzhEfwlpJMUO_886HM-vZeY4ArOsdA9rl4iCCJd0K0JDaXodstN0H.7hlyevU2jOY5kGw Access Passcode: *23rr0^!
	1 Apr	https://nus-sg.zoom.us/rec/share/4bP9EwxftTzQYr_lehPq1c9sTpPkGglhkEYr_ki9vJKVcf3GirhmMopknvfUIRFkY.g_BGLJrt3Ie22dH2 Access Passcode: mbCpj*y9
Week 12, Haldane model, linear response, integer quantum Hall effect, Chern number.	6 Apr	https://nus-sg.zoom.us/rec/share/BkxMf1ZwV9KM3ifnLb_YJSxlohjGWeOut1LS-7ozJBIV0EEjjNCD6xaZDtVZYu9U.kpvlZgowTi8lddsI Access Passcode: *y@5*a+b
	8 Apr	https://nus-sg.zoom.us/rec/share/2zR1hDckKIVQtsw076i5gMTXZK8WlNluR4DQJDhUvOeQTGtcLnJCFBCV3yyDoq-e.vlXc_y0c2Koj0R6H Access Passcode: %7j*5C@b
Week 13, superconductivity, BCS theory	13 Apr	https://nus-sg.zoom.us/rec/share/lamXtETJ132g77FqDeVa_RAumZDvvJQgp_mf1yIPDckjMCBnjI9Ah7fvzVK8cTolk.-bevKwppsNf1EACO Access Passcode: ?cMX5k3u
	15 Apr	https://nus-sg.zoom.us/rec/share/lR4iQnwCZbyQ7UxfjljAXb7OodnsvtqDEnn0GITxYNalCo3XZjKNR0nuGB0mJajJ.dtK9RjhhYMMDjl6Z Access Passcode: m#2D@IVE